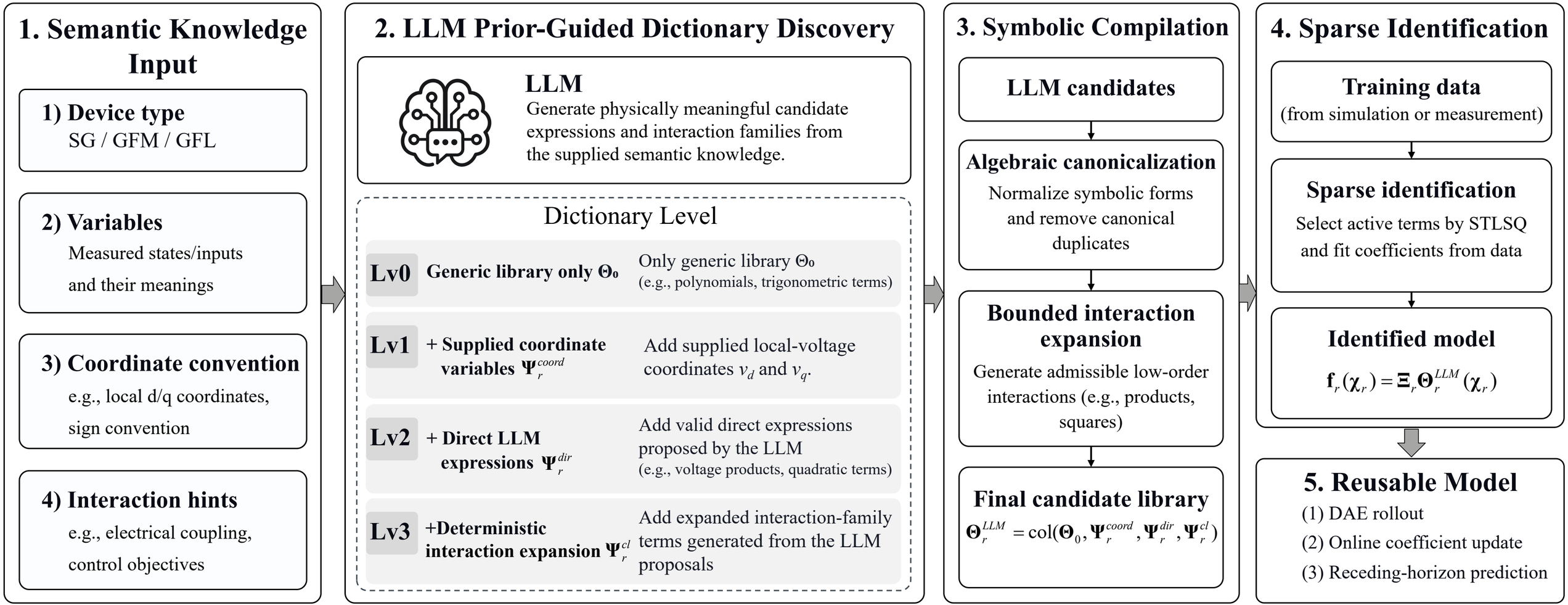

1. Semantic Knowledge Input
1) Device type
SG / GFM / GFL
2) Variables
Measured states/inputs and their meanings
3) Coordinate convention
e.g., local d/q coordinates, sign convention
4) Interaction hints
e.g., electrical coupling, control objectives
2. LLM Prior-Guided Dictionary Discovery
LLM
Generate physically meaningful candidate expressions and interaction families from the supplied semantic knowledge.
Dictionary Level
Lv0 Generic library only Θ0
Only generic library Θ0 (e.g., polynomials, trigonometric terms)
Lv1 + Supplied coordinate variables Ψ_r^coord
Add supplied local-voltage coordinates v_d and v_q.
Lv2 + Direct LLM expressions Ψ_r^dir
Add valid direct expressions proposed by the LLM (e.g., voltage products, quadratic terms)
Lv3 +Deterministic interaction expansion Ψ_r^cl
Add expanded interaction-family terms generated from the LLM proposals
3. Symbolic Compilation
LLM candidates
Algebraic canonicalization
Normalize symbolic forms and remove canonical duplicates
Bounded interaction expansion
Generate admissible low-order interactions (e.g., products, squares)
Final candidate library
Θ_r^LLM = col(Θ_0, Ψ_r^coord, Ψ_r^dir, Ψ_r^cl)
4. Sparse Identification
Training data
(from simulation or measurement)
Sparse identification
Select active terms by STLSQ and fit coefficients from data
Identified model
f_r(χ_r) = Ξ_r Θ_r^LLM(χ_r)
5. Reusable Model
(1) DAE rollout
(2) Online coefficient update
(3) Receding-horizon prediction